
Leonard E. van Dyck

Academic Education

PhD in Psychology (Computational Cognitive Neuroscience)

Justus Liebig University Giessen

- Advisors: Prof. Katharina Dobs & Prof. Martin Hebart

Giessen, Germany

09/2023 - today

MSc in Psychology (Cognitive Neuroscience)

University of Salzburg

- Grade: 1.06 (with Distinction)
- Thesis Advisors: Prof. Mario Braun & Prof. Martin Kronbichler

Salzburg, Austria

10/2020 - 02/2023

BSc in Psychology

University of Salzburg

- Grade: 1.73
- Thesis Advisor: Dr. Walter Gruber

Salzburg, Austria

10/2017 - 07/2020

Work Experience

Guest Researcher

Max Planck Institute for Human Cognitive and Brain Sciences

- Group Leader: Prof. Martin Hebart

Leipzig, Germany

10/2023 – 09/2025

Laboratory Manager

Centre for Cognitive Neuroscience, University of Salzburg

- Method Unit EEG

Salzburg, Austria

10/2020 - 02/2023

Teaching Assistant

Department of Psychology, University of Salzburg

- Bachelor's Seminar "Artificial Intelligence"

Salzburg, Austria

03/2021 - 07/2022

Research Assistant

*Laboratory for Sleep, Cognition, and Consciousness Research,
University of Salzburg*

- Group Leader: Prof. Kerstin Hödlmoser

Salzburg, Austria

03/2020 - 07/2020

Research Internship

Department of Psychology, University of Salzburg

- Clinical and Family Therapy, Prof. Katherine Hertlein (University of Nevada, Las Vegas)

Salzburg, Austria

04/2019 - 07/2019

Grants and Awards

Doctoral Scholarship

10/2023 - today

Studienstiftung des deutschen Volkes e.V.
German Academic Scholarship Foundation

Merit Scholarship

2023

University of Salzburg

ECVP22 Travel Grant

2022

Donders Institute for Brain, Cognition, and Behaviour

Biology Future Prize

2016

Stiftung Natur, Mensch, Kultur

Scientific Contributions

Google Scholar: <https://scholar.google.com/citations?user=5neRr6EAAAAJ>

ORCID: <https://orcid.org/0000-0002-6006-8539>

OA: Open Access

*equal contribution, presenter

Preprints

1. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs., K.* (2025). Multidimensional feature tuning in category-selective areas of human visual cortex. *bioRxiv*.
<https://doi.org/10.1101/2025.06.17.659578> [OA]

Peer-Reviewed Publications

6. **van Dyck, L. E.** & Dobs, K. (2025). Category selectivity as a window into behavioral relevance. *Cognitive Neuroscience*, 1-3. <https://doi.org/10.1080/17588928.2025.2590656> [OA]
5. **van Dyck, L. E.**, Bremmer, F., & Dobs, K. (2024). Artificial intelligence meets body sense: task-driven neural networks reveal computational principles of the proprioceptive pathway. *Signal Transduction and Targeted Therapy*, 9(1), 171.
<https://doi.org/10.1038/s41392-024-01870-9> [OA]
4. **van Dyck, L. E.** & Gruber, W. R. (2022). Modeling biological face recognition with deep convolutional neural networks. *Journal of Cognitive Neuroscience*, 35(10), 1521–1537. https://doi.org/10.1162/jocn_a_02040 [OA]
3. **van Dyck, L. E.**, Denzler, S. J., & Gruber, W. R. (2022). Guiding visual attention in deep convolutional neural networks based on human eye movements. *Frontiers in Neuroscience*, 16.
<https://doi.org/10.3389/fnins.2022.975639> [OA]
2. **van Dyck, L. E.**, Kwitt, R., Denzler, S. J., & Gruber, W. R. (2021). Comparing object recognition in humans and deep convolutional neural networks – An eye tracking study. *Frontiers in Neuroscience*, 15.
<https://doi.org/10.3389/fnins.2021.750639> [OA]
1. Hertlein, K. M. & **van Dyck, L. E.** (2020). Predicting engagement in electronic surveillance in romantic relationships. *Cyberpsychology, Behavior, and Social Networking*, 23(9), 604-610.
<https://doi.org/10.1089/cyber.2019.0424>

Conference Contributions

9. **van Dyck, L. E.** & Dobs., K. (2025). Deep neural networks provide insights into distinct and shared selectivity for faces and bodies in human visual cortex. **Poster** presented at *Cognitive Computational Neuroscience (CCN)*, August 12-15, Amsterdam, NL.
https://2025.ccneuro.org/abstract_pdf/Dyck_2025_Deep_Neural_Networks_Provide_Insights_Distinct.pdf [OA]
8. **van Dyck, L. E.** & Dobs., K. (2025). Deep neural networks provide insights into distinct and shared selectivity for faces and bodies in human visual cortex. **Poster** presented at *Workshop on CONCEPTS, ACTIONS, and OBJECTS (CAOs)*, May 7-9, Rovereto, Italy.
7. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs., K.* (2024). Core neural dimensions of functionally selective areas in the human visual cortex. **Poster** presented at *Neuro-AI-Talks (NEAT)*, September 02-03, Osnabrück, Germany.
6. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs., K.* (2024). Core neural dimensions of functionally selective areas in the human visual cortex. **Poster** presented at *European Conference on Visual Perception (ECVP)*, August 25-29, Aberdeen, Scotland.
5. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs., K.* (2024). Core neural dimensions of functionally selective areas in the human visual cortex. **Poster** presented at *Cognitive Computational Neuroscience (CCN)*, August 6-9, Boston, MA, USA.
https://2024.ccneuro.org/pdf/124_Paper_authored_ManuscriptAuthored.pdf [OA]
4. **van Dyck, L. E.**, Hebart, M. N.*, & Dobs., K.* (2024). Neural representational dimensions capture the nested functional organization of the human visual cortex. **Talk** presented at *SFB Workshop Categorization in Perception and Action: Minds, Models, Mechanisms*, June 30 - July 2, Marburg, Germany.
3. **van Dyck, L. E.**, Dobs., K.*, & Hebart, M. N.* (2024). Data-driven voxel decomposition reveals representational dimensions in functionally selective areas. **Poster** presented at *Workshop on CONCEPTS, ACTIONS, and OBJECTS (CAOs)*, May 9-11, Rovereto, Italy.
2. **van Dyck, L. E.**, Denzler, S. J., & Gruber, W. R. (2022). Analyzing and increasing the similarity of humans and deep convolutional neural networks in object recognition. **Poster** presented at *European Conference on Visual Perception (ECVP)*, August 28 - September 1, Nijmegen, The Netherlands.
1. **van Dyck, L. E.**, Denzler, S. J., Schöllkopf, C. P., & Gruber, W. R. (2022). Spatial similarities between human eye movements and deep convolutional neural network saliency maps across time. **Poster** presented at *Salzburg Mind Brain Annual Meeting (SAMBA)*, July 14-15, Salzburg, Austria.

Theses

- van Dyck, L. E.** (2023). *Unraveling top-down and bottom-up processes in theory of mind with layer fMRI*. [Master's Thesis].
<https://eplus.uni-salzburg.at/obvusbhs/content/titleinfo/8960559> [OA]
- van Dyck, L. E.** & Gruber, W. R. (2020). *Seeing Eye-to-Eye? A comparison of object recognition performance in humans and deep convolutional neural networks under image manipulation*. arXiv [Bachelor's Thesis].
<https://doi.org/10.48550/arxiv.2007.06294> [OA]

Invited Talks

1. *Two Approaches to Increasing the Human-Likeness of Visual Attention in DCNNs*
Vision and Computational Cognition Group (Group Leader: Prof. Martin Hebart)
Max Planck Institute for Human Cognitive and Brain Sciences, September 29, 2022

Science Communication

van Dyck, L. E. & Dobs, K. (2023). Modellierung der biologischen Gesichtswahrnehmung mit Künstlicher Intelligenz. DER AUGENSPIEGEL (German Journal for Ophthalmologists), December 2023, 50-53.

Ad-hoc-Reviewer

The Journal of Neuroscience, PLOS Computational Biology, Frontiers in Neuroscience, British Journal of Psychology, Cognitive Computational Neuroscience Conference

Technical Skills

Programming

- Languages: Python, MATLAB, R, Bash
- Machine Learning: Scikit-learn, PyTorch
- Data Analysis: NumPy, Pandas, Matplotlib, Jupyter
- Version Control: Git

Laboratory

- Experiment Design: PsychoPy, PsychToolbox
- Data Acquisition: EEG, fMRI, Eye Tracking, Psychophysics, Behavior
- Data Analysis: EEG/MEG (FieldTrip), fMRI (SPM, FSL, FreeSurfer, PyCortex), Eye Tracking, Psychophysics, Behavior

Languages

- German (Native)
- English (Proficient - C1)
- French (Intermediate - B1)